

Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int n;
6     int c = 0;
7
8     scanf("%d", &n);
9
10    int i = n / 2;
11    c++;
12
13    while (1)
14    {
15        c++;
16        if (i >= n)
17            break;
18
19        int j = 1;
20
21        while (1)
22        {
23            c++;
24            if (j >= n)
25                break;
26
27            int k = 1;
28
29            while (1)
30            {
31                c++;
32                if (k >= n)
33                    break;
34
35                c++;
36
37                k = k * 2;
38            }
39            j = 2 * j;
40        }
41
42        i++;
43    }
44
45    printf("%d", c);
46
47    return 0;
48 }
```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓